Half Spaces and Distances
Linear Algebra for Machine Learning

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1 Trigonometry Basics

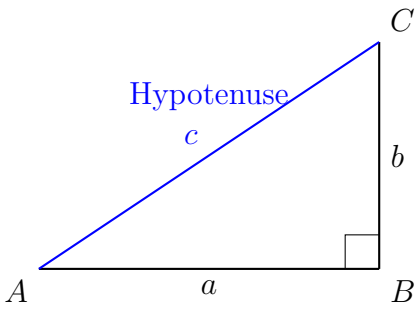
1.1 Pythagorean Theorem

The Pythagorean theorem is fundamental in geometry and essential for understanding distances in machine learning.

Theorem 1.1 (Pythagorean Theorem)**.** In any right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:

$$c^2 = a^2 + b^2 \tag{1}$$

where c is the length of the hypotenuse and a, b are the lengths of the other two sides.



Example 1.1 (Applying Pythagorean Theorem - Finding Hypotenuse)**. Problem:** A right triangle has legs of length 3 and 4. Find the length of the hypotenuse.

Solution:

Using the Pythagorean theorem $c^2 = a^2 + b^2$:

$$c^2 = 3^2 + 4^2 \tag{2}$$
$$c^2 = 9 + 16 \tag{3}$$
$$c^2 = 25 \tag{4}$$
$$c = \sqrt{25} \tag{5}$$
$$c = 5 \tag{6}$$

Answer: The hypotenuse has length 5 units.

Example 1.2 (Finding a Leg Length)**. Problem:** A right triangle has hypotenuse 13 and one leg of length 5. Find the other leg.

Solution:

Let the unknown leg be b. Using $c^2 = a^2 + b^2$:

$$13^2 = 5^2 + b^2 \tag{7}$$
$$169 = 25 + b^2 \tag{8}$$
$$b^2 = 169 - 25 \tag{9}$$
$$b^2 = 144 \tag{10}$$
$$b = \sqrt{144} \tag{11}$$
$$b = 12 \tag{12}$$

Answer: The other leg has length 12 units.

1.2 Trigonometric Ratios

Trigonometric ratios relate the angles of a right triangle to the ratios of its sides.

Definition 1.1 (Basic Trigonometric Ratios)**.** For a right triangle with angle θ:

Sine (sin):

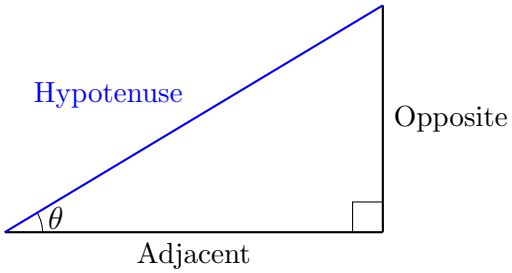
$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}} \tag{13}$$

Cosine (cos):

$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}} \tag{14}$$

Tangent (tan):

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}} \tag{15}$$



Example 1.3 (Computing Trigonometric Ratios)**. Problem:** For a right triangle with legs 3 and 4 and hypotenuse 5, find $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$ where θ is the angle opposite the side of length 3.

Solution:

Identifying the sides relative to angle θ:

- Opposite $= 3$
- Adjacent $= 4$
- Hypotenuse $= 5$

Computing the ratios:

$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{3}{5} = 0.6 \tag{16}$$

$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{4}{5} = 0.8 \tag{17}$$

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{3}{4} = 0.75 \tag{18}$$

Verification using Pythagorean identity:

$$\sin^2(\theta) + \cos^2(\theta) = \left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1 \tag{19}$$

1.3 Relationship Between Slope and Angle

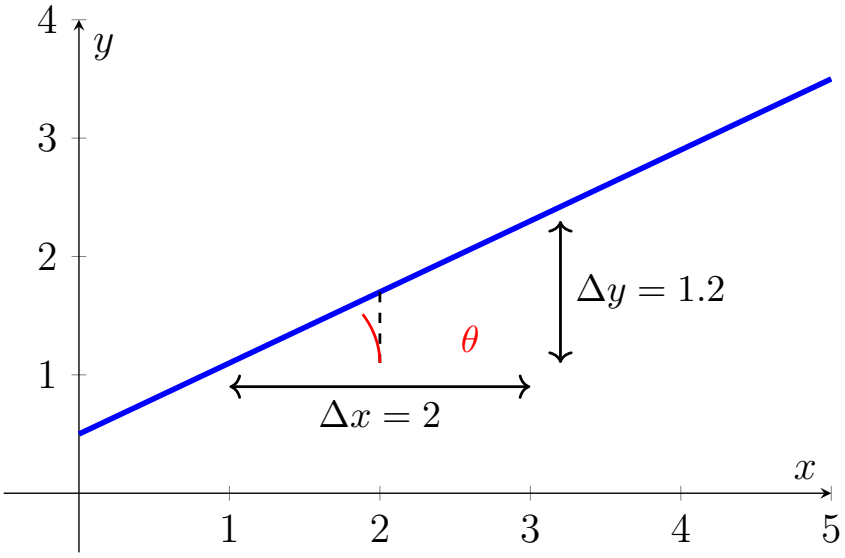
The slope of a line is directly related to the angle it makes with the horizontal axis.

Property 1.1 (Slope-Angle Relationship)**.** For a line making angle θ with the positive x-axis:

$$m = \tan(\theta) \tag{20}$$

Conversely:

$$\theta = \tan^{-1}(m) = \arctan(m) \tag{21}$$



Calculation: $m = \frac{\Delta y}{\Delta x} = \frac{1.2}{2} = 0.6$, and $\theta = \tan^{-1}(0.6) \approx 31$

Example 1.4 (Converting Between Slope and Angle)**. Problem 1:** A line makes a 45 angle with the x-axis. What is its slope?

Solution:

$$m = \tan(45) = 1 \tag{22}$$

Problem 2: A line has slope $m = \sqrt{3}$. What angle does it make with the x-axis?

Solution:

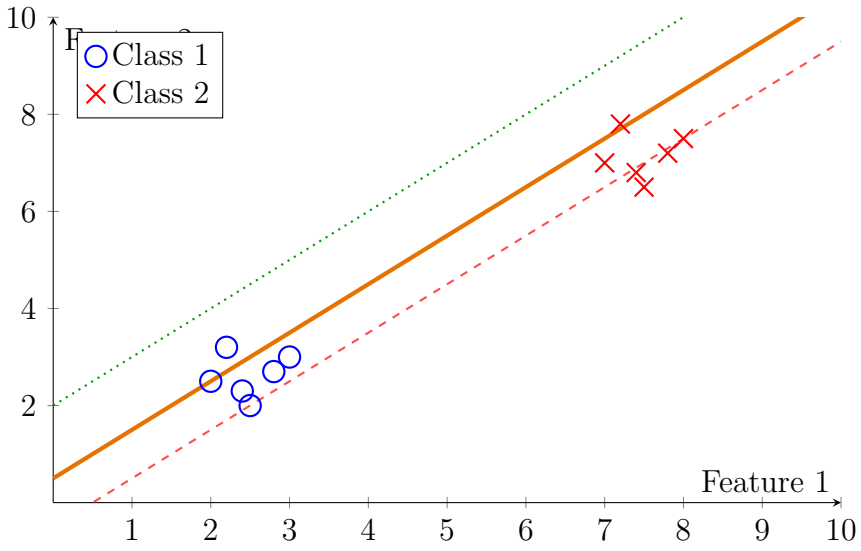
$$\theta = \tan^{-1}(\sqrt{3}) = 60 \tag{23}$$

2 Recap: Classification and Decision Boundaries

2.1 The Classification Problem Revisited

Classification assigns data points to predefined categories using decision boundaries in feature space.

Key Concept: We have data from two classes and need to find a boundary that separates them optimally.



Which boundary is best?

The orange line is optimal because it maintains equal (and maximum) distances from both classes. This principle is called **margin maximization** and is fundamental to Support Vector Machines.

2.2 Why Equal Distances Matter

A decision boundary equidistant from both classes provides:

1. **Safety:** Maximizes the uncertain region where confidence is lowest
2. **Robustness:** Small measurement errors won't easily push points across the boundary
3. **Generalization:** Better classification of new, unseen data

Mathematical Requirement: To find this optimal boundary, we need to:

- Calculate distances from points to hyperplanes
- Understand vector norms and dot products
- Apply perpendicular distance formulas
- Use optimization techniques

3 Forms of Line Equations

Different forms of line equations serve different purposes in analysis and computation.

3.1 Five Standard Forms

Form 1: Slope-Intercept

$$y = mx + c \tag{24}$$

Use: Most intuitive; directly shows slope m and y-intercept c.

Form 2: Point-Slope

$$y - y_1 = m(x - x_1) \tag{25}$$

Use: Convenient when one point (x_1, y_1) and slope m are known.

Form 3: Two-Point

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \tag{26}$$

Use: When two points are known.

Form 4: Intercept

$$\frac{x}{a} + \frac{y}{b} = 1 \tag{27}$$

Use: When x-intercept a and y-intercept b are known.

Form 5: General

$$ax + by + c = 0 \tag{28}$$

Use: Standard form in linear algebra; generalizes to hyperplanes.

3.2 Converting Between Forms

Property 3.1 (General Form to Slope-Intercept)**.** Given $ax + by + c = 0$:

$$by = -ax - c \tag{29}$$

$$y = -\frac{a}{b}x - \frac{c}{b} \tag{30}$$

Therefore:

$$\text{slope} = -\frac{a}{b}, \quad \text{intercept} = -\frac{c}{b} \tag{31}$$

Example 3.1 (Extracting Slope and Intercept)**. Problem:** Find slope and intercept of $3x + 2y + 5 = 0$.

Solution:

Using the formulas:

$$\text{slope} = -\frac{a}{b} = -\frac{3}{2} \tag{32}$$

$$\text{intercept} = -\frac{c}{b} = -\frac{5}{2} \tag{33}$$

Verification by converting to slope-intercept:

$$3x + 2y + 5 = 0 \tag{34}$$

$$2y = -3x - 5 \tag{35}$$

$$y = -\frac{3}{2}x - \frac{5}{2} \tag{36}$$

Answer: Slope $= -\frac{3}{2}$, Intercept $= -\frac{5}{2}$

4 Generalization to Higher Dimensions

4.1 From Lines to Hyperplanes

As we increase the number of features, decision boundaries evolve:

Dimensions	Boundary Type	Equation Form
2D	Line	$w_1x_1 + w_2x_2 + w_0 = 0$
3D	Plane	$w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$
nD	Hyperplane	$\sum_{i=1}^{n} w_ix_i + w_0 = 0$

4.2 Vector Form of Hyperplane

Hyperplane Equation

Expanded Form:

$$w_1x_1 + w_2x_2 + w_3x_3 + \cdots + w_nx_n + w_0 = 0 \tag{37}$$

Vector Form:

$$\vec{w}^T\vec{x} + w_0 = 0 \tag{38}$$

where:

- $\vec{x} = [x_1, x_2, \ldots, x_n]^T \in \mathbb{R}^n$ (feature vector)
- $\vec{w} = [w_1, w_2, \ldots, w_n]^T \in \mathbb{R}^n$ (weight vector / normal vector)
- $w_0 \in \mathbb{R}$ (bias term)

Why Vector Form?

- Compact notation regardless of dimensionality
- Easier algebraic manipulation
- Natural for computer implementation
- Reveals geometric structure (normal vector)

4.3 The Dot Product

Definition 4.1 (Dot Product)**.** For vectors $\vec{a} = [a_1, a_2, \ldots, a_n]^T$ and $\vec{b} = [b_1, b_2, \ldots, b_n]^T$:

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + \cdots + a_nb_n = \sum_{i=1}^{n} a_ib_i \tag{39}$$

Alternative notations: $\vec{a} \cdot \vec{b} = \langle \vec{a}, \vec{b} \rangle = \vec{a}^T\vec{b}$

Example 4.1 (Dot Product Calculation)**. Problem:** Find the dot product of $\vec{a} = [1, 2, 3]$ and $\vec{b} = [-1, 2, 5]$.

Solution:

$$\begin{aligned} \vec{a} \cdot \vec{b} &= (1)(-1) + (2)(2) + (3)(5) && (40) \\ &= -1 + 4 + 15 && (41) \\ &= 18 && (42) \end{aligned}$$

Answer: 18

5 Intersection of Two Lines

5.1 Finding Intersection Points

When two lines intersect, we find the intersection point by solving simultaneous equations.

Example 5.1 (Line Intersection - Addition Method)**. Problem:** Find intersection of $3x - y + 7 = 0$ and $2x + y = 0$.

Solution:
We have:

$$3x - y + 7 = 0 \quad \dots(1) \tag{43}$$
$$2x + y = 0 \quad \dots(2) \tag{44}$$

Step 1: Add equations to eliminate y

$$(3x - y + 7) + (2x + y) = 0 \tag{45}$$
$$5x + 7 = 0 \tag{46}$$
$$x = -\frac{7}{5} = -1.4 \tag{47}$$

Step 2: Substitute into equation (2)

$$2\left(-\frac{7}{5}\right) + y = 0 \tag{48}$$
$$y = \frac{14}{5} = 2.8 \tag{49}$$

Answer: Intersection point is $\left(-\frac{7}{5}, \frac{14}{5}\right)$ or $(-1.4, 2.8)$.
Verification:

$$\text{Equation (1): } 3(-1.4) - 2.8 + 7 = -4.2 - 2.8 + 7 = 0 \tag{50}$$
$$\text{Equation (2): } 2(-1.4) + 2.8 = -2.8 + 2.8 = 0 \tag{51}$$

Example 5.2 (Line Intersection - Substitution Method)**. Problem:** Find intersection of $y = 2x + 7$ and $y = -x + 1$.

Solution:
Set the equations equal:

$$2x + 7 = -x + 1 \tag{52}$$
$$3x = -6 \tag{53}$$
$$x = -2 \tag{54}$$

Substitute back:

$$y = 2(-2) + 7 = 3 \tag{55}$$

Answer: Intersection point is $(-2, 3)$.

6 Distance from Point to Hyperplane

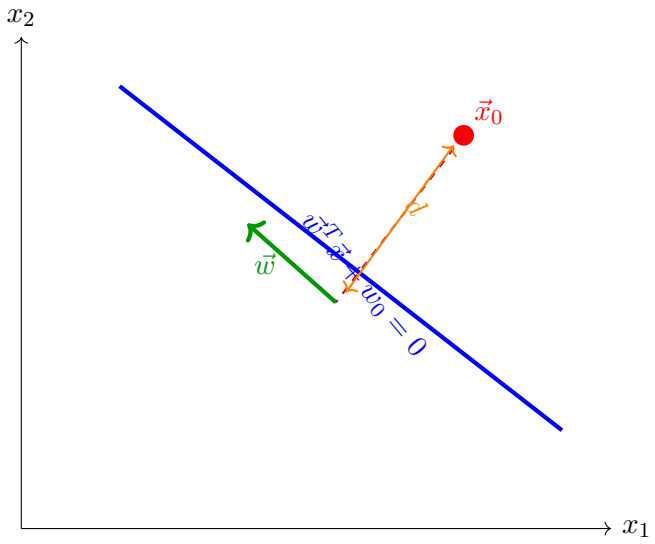
6.1 The Distance Formula

Theorem 6.1 (Point-to-Hyperplane Distance)**.** The distance d from point $\vec{x}_0$ to hyperplane $\vec{w}^T\vec{x} + w_0 = 0$ is:

$$d = \frac{|\vec{w}^T\vec{x}_0 + w_0|}{\|\vec{w}\|} \tag{56}$$

Components:

- Numerator: $|\vec{w}^T\vec{x}_0 + w_0|$ (absolute value of signed distance)
- Denominator: $\|\vec{w}\| = \sqrt{w_1^2 + w_2^2 + \cdots + w_n^2}$ (norm of weight vector)



6.2 Derivation Steps

Key Idea: Distance is measured perpendicular to the hyperplane, along the direction of the normal vector $\vec{w}$.

Step 1: Choose any point $\vec{p}$ on the hyperplane satisfying $\vec{w}^T\vec{p} + w_0 = 0$.
Step 2: Form vector $\vec{v} = \vec{x}_0 - \vec{p}$ from hyperplane to point.
Step 3: Project $\vec{v}$ onto unit normal $\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$.
Step 4: Distance equals projection length:

$$d = \left|(\vec{x}_0 - \vec{p}) \cdot \frac{\vec{w}}{\|\vec{w}\|}\right| = \frac{|\vec{w}^T\vec{x}_0 + w_0|}{\|\vec{w}\|} \tag{57}$$

6.3 Practical Examples

Example 6.1 (Distance in 2D)**. Problem:** Find distance from $(5, 5)$ to line $3x - y + 7 = 0$.
Solution:
Identify components:

- $\vec{x}_0 = [5, 5]^T$
- $\vec{w} = [3, -1]^T$, $w_0 = 7$

Compute numerator:

$$\vec{w}^T\vec{x}_0 + w_0 = 3(5) + (-1)(5) + 7 = 15 - 5 + 7 = 17 \tag{58}$$

Compute denominator:

$$\|\vec{w}\| = \sqrt{3^2 + (-1)^2} = \sqrt{10} \tag{59}$$

Calculate distance:

$$d = \frac{|17|}{\sqrt{10}} = \frac{17\sqrt{10}}{10} \approx 5.38 \tag{60}$$

Answer: $\frac{17\sqrt{10}}{10} \approx 5.38$ units.

Example 6.2 (Distance from Origin)**. Problem:** Find distance from origin to $2x+4y+5=0$.

Solution:

For origin: $\vec{x}_0 = [0,0]^T$

$$\text{Numerator: } 2(0) + 4(0) + 5 = 5 \tag{61}$$

$$\text{Denominator: } \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5} \tag{62}$$

$$d = \frac{5}{2\sqrt{5}} = \frac{5\sqrt{5}}{10} = \frac{\sqrt{5}}{2} \approx 1.12 \tag{63}$$

Answer: $\frac{\sqrt{5}}{2} \approx 1.12$ units.

Example 6.3 (Distance in 3D)**. Problem:** Find distance from $(3,4,5)$ to plane $2x + 4y + 8z + 5 = 0$.

Solution:

Components:

- $\vec{x}_0 = [3,4,5]^T$
- $\vec{w} = [2,4,8]^T$, $w_0 = 5$

Numerator:

$$2(3) + 4(4) + 8(5) + 5 = 6 + 16 + 40 + 5 = 67 \tag{64}$$

Denominator:

$$\sqrt{2^2 + 4^2 + 8^2} = \sqrt{84} = 2\sqrt{21} \tag{65}$$

Distance:

$$d = \frac{67}{2\sqrt{21}} = \frac{67\sqrt{21}}{42} \approx 7.31 \tag{66}$$

Answer: $\frac{67\sqrt{21}}{42} \approx 7.31$ units.

7 Half Spaces

7.1 Definition

A hyperplane divides space into two distinct regions.

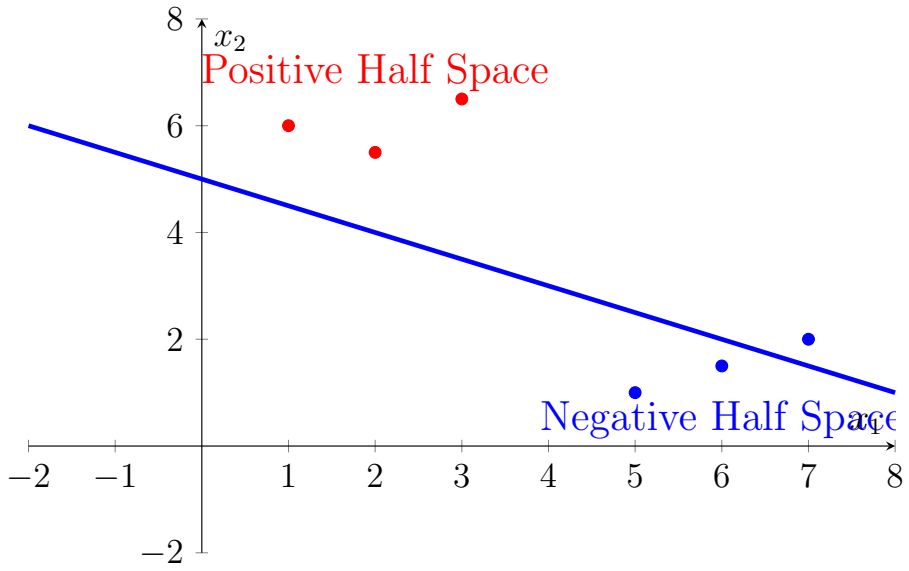
Definition 7.1 (Half Spaces)**.** Given hyperplane $\mathcal{H} : \vec{w}^T\vec{x} + w_0 = 0$:

Positive Half Space:

$$\mathcal{H}^+ = \{\vec{x} : \vec{w}^T\vec{x} + w_0 > 0\} \tag{67}$$

Negative Half Space:

$$\mathcal{H}^- = \{\vec{x} : \vec{w}^T\vec{x} + w_0 < 0\} \tag{68}$$



Example 7.1 (Classifying Points)**. Given:** Hyperplane $2x_1 - 3x_2 + 6 = 0$

Classify: $(0, 0)$, $(3, 4)$, $(0, 2)$

Solution:

For $(0, 0)$:

$$2(0) - 3(0) + 6 = 6 > 0 \quad \text{(Positive half space)} \tag{69}$$

For $(3, 4)$:

$$2(3) - 3(4) + 6 = 0 \quad \text{(On hyperplane)} \tag{70}$$

For $(0, 2)$:

$$2(0) - 3(2) + 6 = 0 \quad \text{(On hyperplane)} \tag{71}$$

8 Comprehensive Practice Problems

Problem 8.1 (Distance Calculations)**.** Find the distance from:

(a) Point $(1, 1)$ to line $x + y - 5 = 0$

(b) Point $(2, 3, 1)$ to plane $x - 2y + 2z - 3 = 0$

(c) Origin to line $4x - 3y + 10 = 0$

Problem 8.2 (Line Intersections)**.** Find intersection points:

(a) $x + 2y = 5$ and $3x - y = 1$

(b) $y = 3x + 2$ and $y = -x + 6$

Problem 8.3 (Half Space Classification)**.** For hyperplane $x_1 + 2x_2 - 4 = 0$, classify:

(a) $(0, 0)$

(b) $(4, 0)$

(c) $(1, 1.5)$